Fine-Tuning CLIP for Zero-Shot Remote Sensing Image Classification via Cross-Dataset Transfer

Vigneshwararaj Sathianarayanan
MSc Computer Science
Linköping University
vigsa504@student.liu.se

Shaurya Thakur
MSc Computer Science
Linköping University
ehath390@student.liu.se

Prithvi Raj Ashokkumar
MSc Computer Science
Linköping University
prias732@student.liu.se

Zeyuan Sheng
MSc Computer Science
Linköping University
zeysh403@student.liu.se

Abstract

We fine-tune OpenCLIP ViT-B/32 [2] with LoRA [3] on the image-caption pairs in RSICD's labelled split, and evaluate on a hidden CodaBench test set whose 21 classes barely overlap RSICD's 31 training classes. Supervised fine-tuning lifts RSICD zero-shot accuracy from 57.18% to 73.28% (macro-F1 0.532→0.732). We then explore two semi-supervised extensions using Split A. Approach 1 adds a SimCLR [8] NT-Xent loss on two augmented views; an ablation over the loss weight $\lambda \in \{0.1, 0.5, 0.9\}$ finds λ=0.1 best at 71.91% accuracy. Approach 1 scores **66.18%** on the leaderboard. Approach 2 adds a novel *SemiCLIP* entropy-minimisation loss over K=31 learnable text anchors initialised from class descriptions, trained jointly with SimCLR under a scheduled λ-warmup over 50 epochs. A matched-conditions ablation reveals a synergy: each auxiliary loss alone hurts, but together they beat supervised-only by $+2.38$ pp. Approach 2 scores **69.89%** on the leaderboard, $+3.89$ pp over the semi supervised baseline.

1 Introduction

Satellite imagery is genuinely odd territory for a vision model trained on the web. Everything is top-down. There is no foreground or background. Objects that dominate the ground such as aircraft, stadiums, highway interchanges appear as tiny clusters of pixels from above. Models pre-trained on ImageNet or web-scraped photos struggle here. The domain gap is not subtle.

CLIP [1] sidesteps the standard classification framing entirely. Rather than predicting a fixed set of class indices, it maps images and text into a shared embedding space. At inference time, you write a text prompt "a satellite image of a forest" and pick whichever class embedding sits closest to the image. No task-specific training on those classes is required. That flexibility matters a lot in this setting.

But the domain gap does not disappear just because CLIP is flexible. Its pre-training corpus is web photos. The image encoder has no idea what a parking lot looks like from an aerial view, or how a runway differs from a river at a pixel level. Fine-tuning on domain-specific image-caption data is the obvious fix and RSICD gives us exactly that. Purpose-built models like GeoRSCLIP [7] outperform general CLIP on aerial imagery, but require large domain-specific corpora we do not have.

This work fine-tunes OpenCLIP ViT-B/32 on RSICD [4] and evaluates on the hidden CodaBench test set, which uses a 21-class taxonomy that barely overlaps with RSICD's 31 training classes.

The model has to use its improved visual understanding to classify images it has never seen, into categories it was never explicitly trained on.

2 Problem Formulation and Datasets

2.1 Datasets

RSICD. The Remote Sensing Image Caption Dataset (RSICD) [4] has 10,921 aerial images covering 31 land-cover categories, each with five human-written captions. Following the project specification, the data splits into three parts:

- *Split A* (8,734 images): unlabelled. Used for unsupervised training if needed; captions exist but are not used here.
- *Split B* (1,094 images): the labelled set. Five captions per image gives 5,470 training pairs. This is what the supervised fine-tuning runs on.
- *Split C* (1,093 images): held out as a local test proxy during development.

Leaderboard test set. The CodaBench leaderboard evaluates against a hidden test set of 2,100 images across 21 land-use categories. These 21 class names are provided in the project specification and form the label space used at inference time.

2.2 Cross-Dataset Transfer Challenge

Here is the thing nobody tells you upfront: RSICD's 31 classes and the leaderboard's 21 target classes come from different sources with different naming conventions. About 7 class names happen to overlap (e.g., *forest*, *beach*, *river*). The other 14 target categories (*baseballdiamond*, *mobilehomepark*, *sparseresidential*, and others) simply do not exist in RSICD. The full breakdown is in Appendix G.

This rules out training a standard classifier on RSICD labels and hoping it transfers. Instead, the model has to learn what aerial imagery looks like through caption training, then generalise that visual knowledge to the 21 target class names at test time. No class-level overlap required, but the image representations have to actually be good.

2.3 Task Definition

Given a test image $\mathbf{x}$ and the 21 target class names $\mathcal{C} = \{c_1, \dots, c_{21}\}$, we predict

$$\hat{y} = \arg \max_{k \in \{1,\dots,21\}} \cos\big(f_I(\mathbf{x}),\ \mathbf{e}_{c_k}\big), \tag{1}$$

where f_I is the L2-normalised image encoder output and $\mathbf{e}_{c_k}$ is the text embedding for class c_k, constructed as described in Section 3.4.

3 Method

3.1 CLIP Architecture

OpenCLIP ViT-B/32 [5] maps images and text into a shared 512-dim unit-sphere embedding space; cosine similarity is a dot product. We start from the `laion2b_s34b_b79k` checkpoint. Architectural details in Appendix B.

3.2 InfoNCE Loss for Supervised Fine-Tuning

The loss is the same one used to train CLIP originally. In a batch of N image-caption pairs from Split B, build an $N \times N$ similarity matrix:

$$S_{ij} = f_I(\mathbf{x}_i)^\top f_T(t_j), \qquad i, j \in \{1, \dots, N\}. \tag{2}$$

Diagonal entries S_{ii} are the correct pairings. Off-diagonal entries are negatives, wrong captions grabbed from the same batch. The loss runs in both directions:

$$\mathcal{L}_{\text{InfoNCE}} \;=\; -\frac{1}{2N}\sum_{i=1}^{N}\left[\;\log\frac{\exp(S_{ii}/\tau)}{\sum_{j=1}^{N}\exp(S_{ij}/\tau)} \;+\; \log\frac{\exp(S_{ii}/\tau)}{\sum_{j=1}^{N}\exp(S_{ji}/\tau)}\;\right], \tag{3}$$

where $\tau = \exp(\texttt{logit\_scale})$ is a learned temperature, initialised from the checkpoint and updated during training. Each image has to score higher against its own caption than against every other caption in the batch, and each caption has to score higher against its own image. The temperature τ controls how sharply the loss penalises confusion between nearby pairs.

3.3 Parameter-Efficient Adaptation with LoRA

Full fine-tuning on 1,094 images would almost certainly damage the pretrained representations. LoRA [3] sidesteps this by keeping the original weights frozen and training small additive updates instead:

$$W' = W + \Delta W = W + \frac{\alpha}{r}BA, \tag{4}$$

where $W \in \mathbb{R}^{d\times k}$ is frozen, $A \in \mathbb{R}^{r\times k}$ and $B \in \mathbb{R}^{d\times r}$ are trainable with rank $r \ll \min(d,k)$, and α scales the update.

We apply LoRA to the MLP projection layers (`c_fc`, `c_proj`) and the attention output projections (`out_proj`) in both transformer stacks, rank $r = 8$, $\alpha = 16$, dropout 0.1. That gives **1,474,560 trainable parameters**: 0.97% of 152.8M total. Everything else stays frozen. Initial experiments using the OpenAI ViT-B-32 checkpoint with full fine-tuning (all parameters updated, no LoRA) peaked at 79.69% on Split C after two epochs but had not stabilised across five epochs. LoRA avoids this instability while also cutting trainable parameters to under 1% of the model.

3.4 Zero-Shot Inference with Multi-Template Averaging

At test time, each of the 21 target class names gets encoded with four different prompt templates: (1) “A satellite image of $\{c\}$.”, (2) “An aerial image of $\{c\}$.”, (3) “A remote sensing image of $\{c\}$.”, (4) “An overhead view of $\{c\}$.”.

The class embedding is the normalised mean of the four:

$$\mathbf{e}_c = \text{normalize}\left(\frac{1}{T}\sum_{t=1}^{T} f_T\big(p_t(c)\big)\right), \quad T = 4. \tag{5}$$

Averaging across templates makes the class embedding less sensitive to any single phrasing [1]. Prediction follows Equation (1).

3.5 Approach 1: Self-Supervised Contrastive Loss on Unlabeled Data

Following SimCLR [8], the loss is computed in the projected space rather than directly on the encoder embeddings, which preserves more general representations. For a batch of M unlabeled images, the 2M projected embeddings are concatenated and the NT-Xent (normalized temperature-scaled cross-entropy) loss is computed as:

$$\mathcal{L}_{\text{unsup}} \;=\; -\frac{1}{2M}\sum_{i=1}^{2M}\log\frac{\exp(\text{sim}(\mathbf{z}_i,\mathbf{z}_{j(i)})/\tau_s)}{\sum_{k=1}^{2M}\mathbf{1}_{[k\neq i]}\exp(\text{sim}(\mathbf{z}_i,\mathbf{z}_k)/\tau_s)} \tag{6}$$

where $\text{sim}(\mathbf{u},\mathbf{v}) = \mathbf{u}^\top\mathbf{v}/\|\mathbf{u}\|\|\mathbf{v}\|$ is the cosine similarity, $j(i)$ denotes the index of the positive pair of sample i (i.e., the other augmented view of the same image), $\mathbf{1}_{[k\neq i]}$ excludes the self-similarity diagonal, and $\tau_s = 0.07$ is a fixed temperature. All $2M-2$ remaining samples within the batch serve as negatives for each anchor, making a larger batch size beneficial for this loss.

The total training loss combines the InfoNCE supervised CLIP contrastive loss $\mathcal{L}_{\text{InfoNCE}}$ on labeled Split B with the self-supervised NT-Xent loss $\mathcal{L}_{\text{unsup}}$ on unlabeled Split A:

$$\boxed{\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{InfoNCE}} + \lambda\cdot\mathcal{L}_{\text{unsup}}} \tag{7}$$

where $\lambda \geq 0$ is a scalar hyperparameter controlling the relative contribution of the unsupervised signal. Setting $\lambda = 0$ recovers the purely supervised objective. The two loss terms operate on different data sources within the same gradient update step: $\mathcal{L}_{\text{InfoNCE}}$ is computed on a batch sampled from Split B, while $\mathcal{L}_{\text{unsup}}$ is computed on a batch sampled from Split A. Since Split A is approximately $8\times$ larger than Split B, the training loop is driven by Split A and Split B is cycled to match.

3.6 Approach 2: SemiCLIP with Learnable Text Anchors

SimCLR pushes views of the same image together but says nothing about class structure. Approach 2 adds a second loss on Split A images that imposes a class-aware prior: SemiCLIP. Let $\mathbf{U} = [\mathbf{u}_1, \ldots, \mathbf{u}_K] \in \mathbb{R}^{K \times d}$ be $K = 31$ learnable anchors, one per RSICD class, initialised from text-encoded descriptive prompts and registered as trainable parameters. For a batch of unlabeled image embeddings $\mathbf{v}_i$ we form soft assignments

$$p_{ik} = \frac{\exp(\mathbf{v}_i^\top \tilde{\mathbf{u}}_k / \tau_a)}{\sum_{k'} \exp(\mathbf{v}_i^\top \tilde{\mathbf{u}}_{k'} / \tau_a)}, \quad \tilde{\mathbf{u}}_k = \mathbf{u}_k / \|\mathbf{u}_k\|, \quad \tau_a = 0.07, \tag{8}$$

and minimise their mean entropy:

$$\mathcal{L}_{\text{semi}} = -\frac{1}{M} \sum_{i=1}^{M} \sum_{k=1}^{K} p_{ik} \log p_{ik}. \tag{9}$$

This pushes each image to commit confidently to one anchor, while the anchors themselves drift to the principal modes of the Split A feature distribution. Because they were initialised from text and stay aligned with the CLIP text encoder during training, the structure they impose is consistent with the captioning loss on Split B.

The full Approach 2 objective combines all three losses with epoch-dependent weights:

$$\mathcal{L}^{(2)} = \mathcal{L}_{\text{InfoNCE}} + \lambda_{\text{sim}}(e)\, \mathcal{L}_{\text{unsup}} + \lambda_{\text{semi}}(e)\, \mathcal{L}_{\text{semi}}, \tag{10}$$

$$\lambda_{\text{sim}}(e) = 1.0 - 0.3e/E_{\max}, \quad \lambda_{\text{semi}}(e) = \begin{cases} 0 & e < 15 \\ \min(0.5,\ 0.5(e-15)/25) & e \geq 15 \end{cases} \tag{11}$$

with $E_{\max} = 50$. SimCLR runs at near-full strength throughout, decaying to 0.7. SemiCLIP stays at zero for 15 epochs to let the anchors settle, then ramps to 0.5 by epoch 40. Activating it earlier collapsed the anchor distribution in our preliminary runs.

3.7 Retrieval

Both retrieval tasks use the joint CLIP embedding space directly [1]. We pre-compute embeddings for all Split C images and their captions, then for each query compute cosine similarity against every candidate and take the top-k. Image-to-text counts a hit if any of the 5 ground-truth captions for that image appears in the top-k; text-to-image counts a hit if the single ground-truth image does. Recall@k is the fraction of queries that hit.

4 Experiments

4.1 Training Setup

AdamW [6], learning rate 10^{-4}, weight decay 0.01, CosineAnnealingLR over 20 epochs ($T_{\max} = 20$), batch size 32. Split B was divided 80/20 into training (4,376 pairs) and validation (1,094 pairs). No gradient clipping. Hardware: Apple Silicon MPS backend, float32 throughout. Full hyperparameter settings are in Table 6.

4.1.1 Approach 1: Semi-Supervised Finetuning

For the semi-supervised finetuning, the training proceeds in two stages. First, a supervised warm-up phase of 3 epochs trains the model exclusively on Split B using supervised loss, allowing LoRA parameters to adapt to the remote sensing domain before the self-supervised signal is introduced. Second, the main semi-supervised phase runs for 20 epochs, with the outer loop driven by Split A

(136 Batches per epoch) and Split B cycled to match. The learning rate follows a cosine annealing schedule over a full 23 epochs and GPU cache is cleared at the end to prevent out of memory error. The learnable temperature is clamped to a maximum value of 100 to prevent numerical instability. We use AdamW optimizer [6] with a cosine learning rate schedule.

4.1.2 Approach 2: Full Three-Loss Pipeline with λ-Warmup

Approach 2 jointly optimises the three losses in Eq. 10 over 50 epochs. No explicit supervised warm-up is needed since SemiCLIP stays at zero for the first 15 epochs by schedule. LoRA uses r=64, α=128 (higher capacity for the extra signals) on the same target layers. A 2-layer MLP projection head (512→512→128) is attached to the image encoder for the SimCLR loss and discarded at inference. AdamW uses two parameter groups: LoRA adapters at 10^{-5}, projection head and anchors at 10^{-4} (weight decay 0.01). The learning rate decays by CosineAnnealingLR over the 50 epochs. Each step samples a Split B batch for InfoNCE and a Split A batch of two augmented views for SimCLR and SemiCLIP; the outer loop is driven by Split A, Split B is cycled. Full hyperparameters in Appendix D.

4.2 Main Results

4.2.1 Supervised Finetuning Results

Table 1 compares the pretrained baseline against the supervised fine-tuned model on the RSICD test split (1,093 images, 31 classes).

Table 1: Top-1 accuracy and macro-F1 on the RSICD test split (1,093 images, 31 classes). Leaderboard accuracy is on the external CodaBench hidden test set (21 classes). The two evaluations use different image sets and class taxonomies and are not directly comparable. †OpenAI ViT-B/32 checkpoint; all other rows use `laion2b_s34b_b79k`.

Model	Training data	RSICD test acc.	RSICD test F1
Pretrained OpenAI(no fine-tuning)†	None	54.25%	0.477
+ Full fine-tuning†	Split B	79.69%	0.782
Pretrained LAION-2B(no fine-tuning)	None	57.18%	0.532
+ LoRA supervised fine-tuning	Split B	**73.28%**	**0.732**
Improvement (LoRA)		+16.10pp	+0.200

Fine-tuning improves RSICD test accuracy by 16.1 percentage points, from 57.18% to 73.28%, with macro-F1 rising from 0.532 to 0.732. The 66% CodaBench score confirms the gain transfers to the hidden external test set, though the gap between local (73.28%) and leaderboard (66%) performance reflects the taxonomy mismatch: locally we evaluate against 31 RSICD class names the model was trained around, while the leaderboard uses 21 different target categories the model has never seen as labels. For reference, the OpenAI ViT-B-32 checkpoint (trained on 400M curated image-text pairs rather than 3.4B LAION pairs) gives a 54.25% zero-shot baseline on the same 1,093-image test set, confirming that the stronger LAION-2B pre-training gives a better foundation before any fine-tuning.

4.2.2 Approach 1 Semi-Supervised Finetuning Results

The semi-supervised finetuned model is evaluated on the RSICD test data (Split-C, 1093 images) for Zero-Shot classification. We perform ablation studies on the hyperparameter lambda on values 0.1, 0.5 and 0.9.

Table 2: Models (with respective lambdas) and their respective Accuracy and F1 score

Model	Zero-shot Accuracy	Macro F1 Score
Semi-Supervised ($\lambda = 0.1$)	71.91%	0.7073
Semi-Supervised ($\lambda = 0.5$)	69.08%	0.6865
Semi-Supervised ($\lambda = 0.9$)	70.27%	0.6985

It is evident from Table 1 that the semi-supervised model with lambda = 0.1, outperforms other models. This is expected because at lambda = 0.1, the loss from unsupervised model has less impact on the total loss compared to lambda = 0.5 and lambda = 0.7. At lambda = 0.5, performance degrades further however, it recovers at lambda = 0.9 indicating that there is no straightforward relation between lambda and zero-shot classification. At lambda = 0.5, the unsupervised signal may interfere with the cross-modal alignment that zero-shot classification depends on, while at lambda = 0.9 the self-supervised signal dominates enough to partially recover visual structure that aids discrimination.

4.2.3 Approach 2 Semi-Supervised Finetuning Results

We ablate the loss components under matched conditions (50 epochs, seed 42, same λ-warmup and cosine LR decay, same data loaders; only the active loss terms change). Accuracy is reported on Split C with the 5-template ensemble. On the leaderboard, the Approach 2 model scores **69.89 %**, $+3.89$ pp above the approach 1 semi supervised baseline (66%).

Table 3: Approach 2 ablation (Split C) and final evaluation. Each auxiliary loss alone hurts; together they beat supervised-only by $+2.38$ pp. Approach 2 trades Split C accuracy for leaderboard accuracy compared with the supervised baseline.

Variant (50 ep, matched)	Acc.	Δ vs Sup.	Δ vs Full
Supervised only	65.23%	—	−2.38
Supervised + SimCLR	61.76%	−3.47	−5.85
Supervised + SemiCLIP	63.59%	−1.64	−4.02
Full (SimCLR + SemiCLIP)	**67.61%**	**+2.38**	—
Supervised baseline vs. Approach 2 (full):			
Split C acc. / macro-F1	73.28% / 0.732		67.61% / 0.663
CodaBench (21 classes, hidden)	NaN		**9.89%**

4.3 Training Dynamics

The supervised baseline's training loss falls from 1.227 to 0.097 over 20 epochs, but validation loss bottoms at epoch 9 (0.607) and rises afterwards: classic overfitting on 4,376 training pairs. We ran all 20 epochs and submitted the final checkpoint; the epoch 9 checkpoint would have been better. Full per-epoch numbers in Appendix F.

4.4 Prompt Template Sensitivity

Individual templates range from 69.08% ("A satellite image of") to 72.92% ("A remote sensing image of"); the 4-template ensemble used throughout beats all of them at 73.28%. "A satellite image of" being the weakest is counter-intuitive but consistent with RSICD captions rarely using that exact phrasing. Per-template numbers in Appendix C.

4.5 Retrieval Evaluation

Table 4 reports image-to-text (I→T) and text-to-image (T→I) recall at $k \in \{1, 5, 10\}$ on Split C (1,093 images, 5,465 captions) for all configurations.

Both semi-supervised approaches beat pretrained CLIP across every cell, confirming the joint embedding space adapted to aerial imagery. Approach 1 (λ=0.1) is strongest on I→T while Approach 2 leads on T→I; for Approach 1, λ matters for I→T (best at 0.1) but the three λ values are within noise for T→I, indicating that text-to-image recall is less sensitive to the degree of unsupervised regularisation.

4.6 Discussion

4.6.1 Supervised Finetuning

The 16-point improvement in zero-shot accuracy on the RSICD test split confirms that contrastive fine-tuning on aerial image captions works. The model learned what aerial textures look like from

Table 4: Retrieval recall on the RSICD Split C. "Pretrained" is the unmodified `laion2b_s34b_b79k` checkpoint. Approach 1 results shown for the best λ=0.1 configuration; full λ breakdown in Appendix E. Values in %.

	Image → Text			Text → Image		
Model	R@1	R@5	R@10	R@1	R@5	R@10
Pretrained CLIP	6.68	17.57	28.09	6.42	21.23	33.78
Supervised fine-tuned	11.25	25.89	36.78	8.23	27.63	41.63
Approach 1 (λ=0.1)	**10.06**	**23.33**	**35.50**	7.41	24.30	38.08
Approach 2 (full)	7.78	21.59	31.38	**7.54**	**25.65**	**39.07**

1,094 images and 5,470 captions, and that knowledge generalised to class names it had never seen as supervised labels.

The template sensitivity results show the ensemble design is justified. The 4.2-point gap between the worst single template (69.08%) and the ensemble (73.28%) is substantial for a system that requires no retraining.

The main failure is overfitting. The best checkpoint was at epoch 9, but we submitted epoch 20. That is fixable with early stopping and no additional compute. The 8,734 unlabelled Split A images are entirely unused here, which is the more interesting gap to close.

4.6.2 Approach 1 Semi-Supervised Finetuning

This ablation study indicates that lower lambda preserves cross-modal alignment better, benefitting both zero-shot classification and Image-to-text retrieval, while text-to-image is largely insensitive to lambda values. The best overall configuration is $\lambda = 0.1$, which achieves the highest accuracy (71.91%), highest macro F1 (0.7073), and best image-to-text R@1 (0.1006). This suggests that the unsupervised loss is most beneficial as a light regularizer rather than a dominant training signal in this low-label regime.

4.6.3 Approach 2 Semi-Supervised Finetuning

The ablation shows a synergy. SimCLR alone hurts (-3.47 pp), SemiCLIP alone hurts (-1.64 pp), but the two together beat supervised-only by $+2.38$ pp. SimCLR enforces augmentation invariance on Split A views and SemiCLIP imposes class-aware structure via the learnable anchors; either signal pulls the encoder away from the captioning loss on its own, but together they act as mutually constraining regularisers and prevent the overfitting the supervised baseline shows at 50 epochs.

The local-vs-leaderboard split is also informative. Approach 2 scores lower than the supervised baseline on Split C (67.61% vs. 73.28%) but higher than semi-supervised approach 1 on the hidden 21-class set (69.89% vs. 66%). The supervised baseline specialises to RSICD's 31 classes; the auxiliary losses trade a bit of that in-distribution sharpness for representations that transfer better to a class space the model never saw. A small in-distribution cost, a larger out-of-distribution gain—the usual regulariser behaviour.

One methodology note. An earlier ablation at 20 epochs without λ-warmup gave the opposite conclusion (everything hurts). The schedule keeps λ_{semi} at zero for the first 15 epochs, so at 20 epochs SemiCLIP is effectively inactive. Matching the ablation to the main training conditions flipped the sign.

4.6.4 Comparison: Approach 1 vs Approach 2

Approach 1 wins on Split C ($+4.30$ pp accuracy) and on image-to-text retrieval; Approach 2 wins on the leaderboard ($+3.89$ pp over baseline) and on text-to-image retrieval. The two are not strictly orderable: the right choice depends on whether the deployment taxonomy is fixed (where Approach 1's lighter regularisation preserves sharpness) or shifts at inference time (where Approach 2's stronger regularisation transfers better).

Table 5: Approach 1 (best λ=0.1) vs Approach 2 (full). Split C numbers are 1,093 images, 31 classes. Best per row in bold.

Metric	Approach 1 (λ=0.1)	Approach 2 (full)
Top-1 accuracy (Split C)	**71.91%**	67.61%
Macro F1 (Split C)	**0.7073**	0.663
I→T R@1 / R@5 / R@10	**10.06 / 23.33 / 35.50%**	7.78 / 21.59 / 31.38%
T→I R@1 / R@5 / R@10	7.41 / 24.30 / 38.08%	**7.54 / 25.65 / 39.07%**
CodaBench leaderboard (21 cls)	66.18%	**69.89%**
Epochs / LoRA rank	23 / 8	50 / 64
LR schedule	Cosine (23 ep)	Cosine (50 ep)
λ schedule	Fixed (λ=0.1)	Eq. 11

5 Conclusion

Supervised LoRA fine-tuning lifts Split C accuracy from 57.18% to 73.28%. Approach 1 (InfoNCE + SimCLR, λ=0.1) gives a small lift on Split C (71.91%) and confirms that the unsupervised signal works best as a weak regulariser (66% on CodaBench). Approach 2 adds SemiCLIP with learnable text anchors and a scheduled λ-warmup over 50 epochs; the ablation shows a synergy neither single-loss variant has, and threaches **69.89 %** on CodaBench, $+3.89$ pp over the supervised baseline. The Split C/leaderboard split is informative: Approach 1 wins when the taxonomy is fixed, Approach 2 wins when it shifts.

Limitations. 1,094 training images is not much; early stopping would have been free performance for the supervised baseline. Approach 2 ablations used a single seed each, so $\pm$1–2 pp variance is not characterised. The Split C/leaderboard gap is structural—closing it would need supervision on the 21 target classes.

6 Ethical Considerations

Land-use classification from satellite imagery is useful. Urban planners, environmental researchers, and disaster response teams all have legitimate reasons to want it. But the same model that identifies forests also identifies military installations, refugee camps, and large gatherings. That dual-use problem is real and not hypothetical. It is the normal state of remote sensing technology.

We used publicly available datasets with no personal information in them. We are not releasing model weights. The LAION-2B pre-training corpus has known issues around scraped web content; consult the OpenCLIP and LAION documentation for specifics. For an academic project the risk is low. A deployed system would need proper answers to these questions.

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Author Contributions

Prithvi Raj Ashokkumar designed and implemented the data preprocessing pipeline, covering dataset parsing, split construction, class-label mapping, and PyTorch dataloaders for all three RSICD splits. Designed and implemented the zero-shot baseline evaluation framework including the 4-template ensemble inference, establishing a 54.25% baseline on the 31-class RSICD test split. Conducted full fine-tuning experiments that peaked at 79.69% accuracy, directly informing the team's decision to adopt LoRA for the main pipeline. Responsible for report writing and documentation of correspoding parts.

Vigneshwararaj Sathianarayanan designed and implemented Approach 2, the full three-loss semi-supervised pipeline. This includes the SemiCLIP entropy-minimisation loss over $K=31$ learnable surrogate text anchors, the projection head used for the SimCLR objective, the combined three-term loss with the epoch-dependent λ-warmup schedule, the 50-epoch training infrastructure with two-group AdamW and cosine learning-rate annealing, the matched-conditions ablation study, and the cross-modal retrieval evaluation with the pretrained-baseline comparison. Also drafted the corresponding method, training-setup, results, and discussion subsections.

Shaurya Thakur designed and implemented the Approach1 full semi-supervised training pipeline. This included building the unlabeled Split A dataset class with a two-view augmentation pipeline tailored to remote sensing imagery, incorporating vertical flipping as a domain-appropriate augmentation given the lack of canonical orientation in aerial images. Implemented the SimCLR-style NT-Xent self-supervised contrastive loss with a non-linear projection head mapping. Designed the two-stage training procedure consisting of a 3-epoch supervised warm-up phase on Split B followed by 20 epochs of joint semi-supervised training driven by Split A, with Split B cycled to match.

Zeyuan Sheng contributed to experimental evaluation, comparative analysis, and interpretation of model behavior across different training settings. Played an important role in organizing and benchmarking results for zero-shot classification and cross-modal retrieval tasks, and in analyzing performance trends across supervised and semi-supervised approaches. Also contributed to the discussion section by refining the interpretation of experimental findings and ensuring consistency between quantitative results and conclusions. In addition, supported the final integration and polishing of the report.

AI Tool Declaration

Claude (Anthropic) was used for code debugging, conceptual discussion, and drafting. All experiments were run by the authors. All reported numbers are from our own runs.

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A Hyperparameter Summary

Table 6: Training hyperparameters for supervised fine-tuning.

Hyperparameter	Value
Base model	OpenCLIP ViT-B/32 (`laion2b_s34b_b79k`)
LoRA rank r	8
LoRA scale α	16
LoRA target layers	`c_fc`, `c_proj`, `out_proj`
LoRA dropout	0.1
Trainable parameters	1,474,560 of 152,751,873 (0.97%)
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	0.01
LR schedule	CosineAnnealingLR ($T_{\max} = 20$)
Training epochs	20
Batch size	32
Training pairs	4,376 (80% of Split B $\times$ 5 captions)
Validation pairs	1,094 (20% of Split B $\times$ 5 captions)
Inference templates	4
Hardware	RTX Ada 4000
Approach 1 Lambda	0.1, 0.5 and 0.9
Approach 2 λ schedule	Eq. 11

B OpenCLIP ViT-B/32 Architecture

The image encoder splits each 224×224 image into 7×7=49 non-overlapping 32×32 patches, projects each to 768 dimensions, prepends a learnable `[CLS]` token, adds positional embeddings, and runs the sequence through 12 self-attention layers. The final `[CLS]` output is projected to 512 dimensions and L2-normalised. The text encoder is a 12-layer causal transformer (GPT-style) with BPE tokenisation (max 77 tokens); the `[EOS]` representation is projected to 512 and normalised. Both outputs land on the same unit hypersphere.

C Prompt Template Breakdown

Table 7: Per-template accuracy on the RSICD test split (supervised model). The ensemble averages the four template embeddings.

Template	Accuracy	Macro-F1
"A satellite image of {c}."	69.08%	0.688
"An aerial image of {c}."	71.82%	0.718
"A remote sensing image of {c}."	72.92%	0.719
"An overhead view of {c}."	72.28%	0.719
4-template ensemble	**73.28%**	**0.732**

D Approach 2 Hyperparameters

Table 8: Approach 2 hyperparameters. Differences from the supervised baseline (Table 6) are LoRA capacity, the additional losses and their λ-schedule, two-group optimiser, and 50 epochs.

Hyperparameter	Value
Base model	OpenCLIP ViT-B/32 (`laion2b_s34b_b79k`)
LoRA rank r / scale α / dropout	64 / 128 / 0.1
LoRA target layers	`c_fc`, `c_proj`, `out_proj`
Projection head (SimCLR)	MLP $512 \to 512 \to 128$, ReLU, BN
Surrogate text anchors (SemiCLIP)	K=31 learnable, init. from class descriptions
LR (LoRA params) / LR (head + anchors)	1×10^{-5} / 1×10^{-4}
Optimiser	AdamW, weight decay 0.01
LR schedule	CosineAnnealingLR ($T_{\max} = 50$, $\eta_{\min} = 10^{-7}$)
Epochs	50
SimCLR temperature τ_s / SemiCLIP τ_a	0.07 / 0.07
λ-schedule	Eq. 11

E Approach 1 Retrieval by λ

Table 9: Approach 1 retrieval on Split C, all three λ values. Image-to-text retrieval is sensitive to λ (best at 0.1); text-to-image is essentially flat across the three values.

		Image → Text			Text → Image		
Model	λ	R@1	R@5	R@10	R@1	R@5	R@10
Approach 1	0.1	0.1006	0.2333	0.3550	0.0741	0.2430	0.3808
Approach 1	0.5	0.0924	0.2306	0.3532	0.0712	0.2487	0.3832
Approach 1	0.9	0.0970	0.2123	0.3038	0.0714	0.2474	0.3833

F Supervised Baseline Loss Curve

Table 10: InfoNCE training and validation loss across all 20 epochs of the supervised baseline. Validation loss bottoms at epoch 9 (0.607).

Epoch	Train	Val	Epoch	Train	Val
1	1.227	0.845	11	0.160	0.633
2	0.733	0.729	12	0.151	0.632
3	0.556	0.686	13	0.139	0.613
4	0.454	0.642	14	0.129	0.618
5	0.368	0.635	15	0.126	0.616
6	0.313	0.628	16	0.113	0.616
7	0.267	0.617	17	0.111	0.622
8	0.231	0.611	18	0.103	0.626
9	0.201	**0.607**	19	0.108	0.625
10	0.192	0.631	20	0.097	0.624

G Target Class List and RSICD Overlap

Table 11 lists all 21 target classes used at inference. Seven (†) share their exact name with an RSICD training class. The model has to generalise to the remaining 14 purely from its aerial image representations.

Table 11: The 21 target classes used for zero-shot inference. † marks a direct name match with RSICD's 31-class taxonomy.

agricultural	airplane	baseballdiamond	beach†
buildings†	chaparral	denseresidential	forest†
freeway†	golfcourse	harbor	intersection
mediumresidential	mobilehomepark	overpass	parkinglot†
river†	runway†	sparseresidential	storagetanks
tenniscourt			